Predictive Operator Theory for Observable Dynamical Systems

Patrick S. Mahon

Abstract

This paper develops the operator theory of predictive dynamics arising from observable experiments. The predictive factorization theorem of the preceding paper forces conditional expectations of future observables to factor through a minimal predictive state space Q_*. This paper shows that predictive evolution on O_{Q_*} is represented by a strongly continuous semigroup of Markov operators $\{K_t\}_{t\geq 0}$, studies its infinitesimal generator, and situates the construction within classical Koopman and Markov operator theory.

1 Introduction

This paper studies the dynamical structure that emerges from prediction in observable systems.

The observational program of the preceding works establishes the chain

$$\text{observable experiments} \;\to\; \text{probability} \;\to\; \text{predictive state} \;\to\; \text{operator dynamics.}$$

The preceding papers construct the canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$ and the minimal predictive query $Q_* : \Omega \to O_{Q_*}$ whose states correspond exactly to distinct predictive laws. Because the predictive law factors through Q_*, conditional expectations of future observables define operators acting on functions of the predictive state; the present paper is a study of those operators and their dynamical structure.

For each prediction horizon $t \geq 0$, define the predictive operator by

$$(K_t g)(q) = E[g(F_t) \mid Q_* = q].$$

The central result (Theorem 1) is that the family $\{K_t\}_{t\geq 0}$ forms a Markov operator semigroup on O_{Q_*}, from which an infinitesimal generator and predictive Kolmogorov equation follow by standard C_0-semigroup theory. Figure 1 summarises the derivational structure of the paper.

Structure of the paper. Section 2 fixes the standing framework. Section 3 defines the predictive operators and establishes basic positivity properties. Section 4 proves the semigroup theorem and the Markov kernel representation. Section 5 defines the infinitesimal generator and derives the predictive Kolmogorov equation. Section 6 situates the construction within classical Koopman and Markov operator theory. Sections 7–9 address spectral structure, the observable dynamical system, and empirical approximation.

2 Standing framework

The following objects are assumed throughout.

- A query system $(\mathcal{Q}, \preceq)$ and canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$.
- The minimal predictive query $Q_* : \Omega \to O_{Q_*}$, whose states correspond exactly to distinct predictive laws.
- A parametrized family of future observables $F_t : \Omega \to O_F$, $t \geq 0$, with O_F standard Borel.

Predictive state space $Q_* : \Omega \to O_{Q_*}$ (from Paper 2)

$\downarrow$

Operators $(K_t g)(q) = E[g(F_t) \mid Q_* = q]$

$\downarrow$

Semigroup $K_{t+s} = K_t K_s, \quad K_0 = \mathrm{Id}$

$\downarrow$

Markov kernel $(K_t g)(q_*) = \int g \, dP_t(q_*, \cdot)$

$\downarrow$

Generator $Ag = \lim_{t \to 0}(K_t g - g)/t$

$\downarrow$

Kolmogorov equation $\partial_t u = Au, \quad u(0, \cdot) = g$

Figure 1: Derivational structure of Paper 3.

- A family of Markov kernels $\{P_t\}_{t \geq 0}$ on O_{Q_*} representing the predictive transition laws, satisfying the Chapman–Kolmogorov equation $P_{t+s} = P_t \circ P_s$.

Observable functions g are taken to be bounded and measurable. Boundedness ensures integrability under every predictive law and allows semigroup compositions and duality integrals to be handled by Bochner integration without additional hypotheses.

Definition 1 (Predictive kernel family)**.** *A* predictive kernel family *on O_{Q_*} is a collection of Markov kernels $\{P_t\}_{t \in T}$ ($T = \mathbb{N}$ or $[0, \infty)$) satisfying:*

1. Identity at zero*: $P_0(q_*, \cdot) = \delta_{q_*}$,*
2. Markov property*: each $P_t(\cdot, \cdot)$ is a probability kernel on O_{Q_*},*
3. Chapman–Kolmogorov*: $P_{t+s}(q_*, \cdot) = \int P_t(q', \cdot) \, P_s(q_*, dq')$ for all $t, s \in T$.*

Remark 1 (Lean formalization)**.** *The Lean formalization uses a structure `PredictiveSemigroupKernel` with exactly these three fields. The discrete-time case $T = \mathbb{N}$ is fully formalized; the continuous-time case $T = [0, \infty)$ and the associated C_0-semigroup theory are not yet formalized.*

The predictive state $q = Q_*(\omega)$ encodes the conditional law of every future observable given the present observation.

3 Predictive operators

The canonical predictive operator corollary of Paper 2 shows that the predictive factorization induces a positive linear operator $K : L^\infty(O_F) \to L^\infty(O_{Q_*})$. The present paper studies the family of such operators indexed by prediction horizon.

Definition 2 (Predictive operator)**.** *For $t \geq 0$ and bounded measurable $g : O_F \to \mathbb{R}$, define*

$$(K_t g)(q) = E[g(F_t) \mid Q_* = q].$$

Proposition 1. *For bounded measurable g, K_t is a positive linear operator that preserves constants.*

4 Operator semigroup

Prediction at successive horizons induces a family $\{K_t\}_{t\geq 0}$ of operators on $L^\infty(O_{Q_*})$. The following theorem shows that this family forms a semigroup.

Theorem 1 (Semigroup property)**.** *Assume the Markov kernels $\{P_t\}$ satisfy the Chapman–Kolmogorov equation*

$$P_{t+s}(q_*, \cdot) = \int P_t(q', \cdot)\, P_s(q_*, dq').$$

Then for every bounded measurable $g : O_{Q_*} \to \mathbb{R}$,

$$K_{t+s} g = K_t(K_s g), \qquad t, s \geq 0.$$

Each operator K_t is positive, preserves constants, and $K_0 = \mathrm{Id}$.

Proof. Fix q_* and compute:

$$\begin{aligned}
(K_{t+s} g)(q_*) &= \int g(q')\, P_{t+s}(q_*, dq') \\
&= \int g(q') \int P_t(q'', dq')\, P_s(q_*, dq'') \\
&= \int \left(\int g(q')\, P_t(q'', dq') \right) P_s(q_*, dq'') \\
&= \int (K_t g)(q'')\, P_s(q_*, dq'') = (K_t(K_s g))(q_*).
\end{aligned}$$

The interchange of integration is justified by Fubini's theorem for Bochner integrals against the kernel composition (bounded g ensures integrability under every finite kernel measure). □

Remark 2 (Formalization scope)**.** *The Lean formalization of the semigroup property (`semigroup_property`) covers the discrete-time case $t, s \in \mathbb{N}$ for bounded measurable observables. The explicit uniform bound $\exists C,\ \forall q,\ |g(q)| \leq C$ is a Lean hypothesis, required for the Bochner–Fubini step (`Kernel.integral_comp`) used in the kernel composition argument. The continuous-time version ($t, s \in [0, \infty)$) and the associated C_0-semigroup theory, including the infinitesimal generator (Section 5), are not yet formalized.*

Proposition 2 (Markov kernel representation)**.** *Since $O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F)$ embeds into the standard Borel space $\mathcal{P}(O_F)$ (which is standard Borel when O_F is), the disintegration theorem applies and there exists a Markov kernel $P_t(q_*, \cdot)$ on O_{Q_*} such that*

$$(K_t g)(q_*) = \int g(f)\, P_t(q_*, df).$$

Thus each K_t is a Markov operator on O_{Q_*}.

Remark 3 (Predictive states as a minimal Markov completion)**.** *The predictive factorization theorem implies that the future law of F_t depends on the present only through Q_*:*

$$P(F_t \in A \mid \Omega) = P(F_t \in A \mid Q_*).$$

Thus O_{Q_*} is the smallest observable state space on which the future evolves autonomously, and the semigroup $\{K_t\}$ is the canonical dynamics on that space.

5 Predictive generator

The semigroup $\{K_t\}$ admits an infinitesimal generator in the standard sense of C_0-semigroup theory.

Definition 3 (Predictive generator)**.** *For functions g in the domain*

$$\mathcal{D}(A) = \left\{ g : \lim_{t\to 0} \frac{K_t g - g}{t} \text{ exists in } L^\infty(O_{Q_*}) \right\},$$

define the predictive generator

$$Ag = \lim_{t\to 0} \frac{K_t g - g}{t}.$$

Theorem 2 (Existence of the predictive generator)**.** *If $\{K_t\}_{t\geq 0}$ is strongly continuous on a Banach space of bounded observable functions on O_{Q_*}, then A is a densely defined closed operator.*

Theorem 3 (Predictive Kolmogorov equation)**.** *For every $g \in \mathcal{D}(A)$, the function $u(t,q) = (K_t g)(q)$ satisfies*

$$\partial_t u(t,q) = Au(t,q), \qquad u(0,q) = g(q).$$

Remark 4 (Predictive evolution equation)**.** *The generator A captures instantaneous predictive evolution. In the comparison section below it is shown that A reduces to the Koopman generator in deterministic systems and to the infinitesimal generator of the associated Markov process in the stochastic setting.*

6 Relation to Koopman and transfer operators

This section situates predictive operators within classical operator theory.

Proposition 3 (Deterministic specialization)**.** *If the predictive kernel is deterministic,*

$$F_t = \phi_t(Q_*),$$

then

$$(K_t g)(q) = g(\phi_t(q)).$$

Thus predictive operators reduce to the classical Koopman semigroup.

Remark 5 (Flow semigroup law and point separation)**.** *In the deterministic setting the Chapman–Kolmogorov equation implies the flow law $\phi_{t+s} = \phi_t \circ \phi_s$. The Lean formalization (`deterministic_semigroup`) derives this via the injectivity of the Dirac embedding $x \mapsto \delta_x$: from $\delta_{\phi_{t+s}(q)} = \delta_{\phi_t(\phi_s(q))}$ one concludes $\phi_{t+s}(q) = \phi_t(\phi_s(q))$. This step requires that the observable state space O_{Q_*} separates points in the sense that distinct points are separated by measurable functions (i.e. the instance `[SeparatesPoints` O_{Q_*}`]` in Lean). This condition is satisfied by all standard Borel spaces and all Hausdorff spaces carrying a separating family of continuous functions.*

Proposition 4 (Generator in deterministic systems)**.** *If the predictive operators arise from a deterministic flow $q_t = \phi_t(q)$, then*

$$Ag = \nabla g \cdot v,$$

where v is the vector field generating the flow.

Proposition 5 (Generator in stochastic systems)**.** *If the predictive dynamics form a Markov process, the predictive generator A coincides with the infinitesimal generator of the associated Markov process.*

Proposition 6 (Dual operator and Koopman–Perron duality)**.** *Let $P_t(q_*, \cdot)$ be the Markov kernel of Proposition 2. Let μ be a finite measure on O_{Q_*} and let $g : O_{Q_*} \to \mathbb{R}$ be a bounded measurable function. Define*

$$(P_t^*\mu)(A) = \int P_t(q_*, A)\, \mu(dq_*).$$

Then

$$\int (K_t g)(q_*)\, \mu(dq_*) = \int g(q_*)\, (P_t^*\mu)(dq_*).$$

Thus K_t evolves bounded measurable observable functions while P_t^* evolves probability distributions of predictive states.

Proof. By Fubini's theorem for Bochner integrals against kernel compositions (boundedness of g and finiteness of μ ensure all integrands are integrable):

$$\int (K_t g)(q_*)\, \mu(dq_*) = \int\int g(q')\, P_t(q_*, dq')\, \mu(dq_*) = \int g(q')\, (P_t^*\mu)(dq').$$

$\square$

Remark 6 (Koopman–Perron duality from prediction)**.** *The duality between K_t and P_t^* is a direct consequence of the predictive factorization: once the predictive law factors through Q_*, Markov operators on O_{Q_*} and their duals on predictive distributions arise automatically. Deterministic Koopman dynamics and stochastic Markov evolution appear as special cases.*

7 Spectral structure

Eigenfunctions of the predictive operators characterize coherent predictive observables and long-term predictive structure.

Definition 4 (Predictive eigenfunction)**.** *A measurable function $g : O_{Q_*} \to \mathbb{R}$ is a predictive eigenfunction with eigenvalue λ_t if*

$$K_t g = \lambda_t g.$$

8 Observable dynamical systems

The pair $(O_{Q_*}, \{K_t\}_{t\geq 0})$ defines the observable dynamical system associated with the predictive experiment: an intrinsic description of the dynamics expressed entirely in terms of the predictive state space and its operator evolution.

9 Empirical approximation

This section studies approximation of predictive operators from observable time-series data.

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